

TYAI Assignment

Q1. Text Processing & Linguistic Foundations (Module I)

- a) Define Natural Language Processing (NLP). Explain its scope and discuss any four real-world applications.
- b) Explain the complete NLP Pipeline with a neat block diagram and describe the purpose of each stage.
- c) Explain the following text preprocessing techniques with suitable examples:
- Tokenization
 - Stopword Removal
 - Text Normalization (Lowercasing and Punctuation Removal)
- d) Differentiate between Stemming and Lemmatization. Explain the Porter Stemmer and WordNet Lemmatizer with suitable examples.

Q2. Text Representation Techniques (Module II)

- a) Explain the Bag of Words (BoW) model. Construct the BoW matrix for the following corpus:
- Machine learning improves prediction.
 - Deep learning uses neural networks.
 - Machine learning is powerful.
- b) Explain Term Frequency (TF) and Inverse Document Frequency (IDF). Calculate the TF-IDF score for any one term using the above corpus.
- c) Explain Word2Vec in detail. Compare the CBOW and Skip-Gram architectures with neat diagrams.
- d) Compare the following text representation techniques based on representation, advantages, limitations, and applications:
- One-Hot Encoding
 - Bag of Words
 - TF-IDF
 - Word2Vec

Q3. NLP Models & Algorithms (Module III)

- a) Explain the Naive Bayes Classifier with its assumptions and discuss its applications in text classification.**
- b) Explain the Hidden Markov Model (HMM) and describe how it is used for Part-of-Speech (POS) Tagging.**
- c) Explain Language Models. Differentiate between Unigram, Bigram, and Trigram models with suitable examples.**
- d) Explain the following evaluation metrics with mathematical formulas and suitable examples:**
 - Accuracy
 - Precision
 - Recall
 - F1-Score

Q4. Deep Learning in NLP (Module IV)

- a) Explain the architecture and working of a Feedforward Neural Network and a Recurrent Neural Network (RNN). Compare both models.**
- b) Explain the architecture of an LSTM network. Describe the functions of the Forget Gate, Input Gate, and Output Gate with a neat diagram.**
- c) Explain the Attention Mechanism and discuss its importance in sequence modeling.**
- d) Explain the Transformer Architecture. Describe the concepts of Encoder–Decoder Architecture and Self-Attention with neat diagrams. Also compare RNN, LSTM, and Transformer models.**